

Bridging Policy and Practice: A Literature Review on Digital Cultural Mapping for Rural Development in the Philippines

Methodologies for Digital Cultural Mapping in the Philippine Context

The development of an Interactive Digital Cultural Map and Information System for Mangatarem, Pangasinan, is situated within a growing body of work focused on leveraging technology for cultural documentation and local development. An examination of recent literature reveals that digital cultural mapping in the Philippines is not merely a technical exercise but a structured process grounded in national policy, institutional frameworks, and participatory methodologies. This foundation provides a robust blueprint for the proposed project, outlining both established precedents and critical challenges that must be navigated for successful implementation. The core methodology emerging from the literature is Participatory Geographic Information Systems (PGIS), which emphasizes community engagement over top-down data acquisition, ensuring that the resulting map reflects authentic local values and knowledge [277278](#). This approach is particularly salient for a municipality like Mangatarem, where the goal is not only to document heritage but also to foster a sense of ownership and pride among its residents.

The conceptualization of cultural mapping in the Philippines is strongly supported by national-level mandates and institutional actors. The National Commission for Culture and the Arts (NCCA) plays a pivotal role in guiding and supporting cultural mapping projects across the country, working in collaboration with Local Government Units (LGUs) [32](#) [150](#). The legal framework for this work is often rooted in laws that recognize the importance of preserving the nation's cultural assets, providing a clear justification for municipal-level initiatives [32](#). LGUs are recognized as the primary implementers of these programs at the grassroots level, tasked with identifying, documenting, and promoting their unique cultural landscapes [275](#). The Department of Tourism (DOT) further reinforces this by acknowledging the crucial role of cultural mapping in developing sustainable tourism products and destinations [247](#). This alignment between national policy, institutional support, and sectoral strategy creates a fertile environment for a project like

the one proposed for Mangatarem. It is not an isolated endeavor but part of a larger national movement towards recognizing and valorizing local culture through systematic documentation.

Several recent studies have provided concrete examples of how cultural mapping has been successfully implemented at the municipal and provincial levels, offering direct precedents for the Mangatarem project. A case study from San Pablo City detailed the process of creating a cultural map of its built heritages, aiming for greater accuracy and comprehensiveness [83](#). Similarly, a project in Guimaras province resulted in a comprehensive culture map that chronicled the events shaping the island province's identity [85](#). These examples demonstrate that the process involves more than just cartography; it requires historical research, community consultation, and strategic planning to create a resource that serves both administrative and promotional purposes. The Municipality of Narra, for instance, utilized its cultural map to support the functions of its Municipal Tourism Development and Promotions Office (MTDPO), showing a direct link between the mapping initiative and economic development goals [73](#). The experience of the Ifugao Rice Terraces, where a heritage cluster was integrated into the municipality's Comprehensive Land-use Plan (CLUP), highlights the potential for cultural mapping to influence long-term land use and conservation strategies [273](#). For Mangatarem, these precedents suggest that the digital map can serve multiple functions: as a tool for heritage conservation, a guide for sustainable tourism planning, and a reference for educational and community-building activities. The literature suggests that the scope of such a map should encompass not only tangible assets like historical buildings and archaeological sites but also intangible aspects like traditions and community practices, which are often documented through cultural mapping projects [122](#).

The dominant methodological approach identified in the literature is Participatory GIS (PGIS). PGIS is defined as an approach that combines traditional GIS technologies with participatory learning and action methods, empowering communities to become active participants in the mapping process [277](#). This is a significant departure from conventional top-down mapping, which relies solely on expert-driven data collection. The literature consistently emphasizes that involving local stakeholders—including officials, elders, artists, and residents—is critical to the success and legitimacy of any cultural mapping project [42](#) [207](#). By partnering with the community, researchers and planners can tap into local knowledge that may not be recorded in official archives, leading to a more nuanced and accurate representation of the cultural landscape [144](#). Studies have demonstrated the application of PGIS for a wide range of purposes relevant to Mangatarem, including the identification of tangible and intangible cultural heritage, community-based disaster management, and sustainable land use planning [100110254](#). For example, a study on

Vigan's heritage conservation highlighted the city's effective use of cultural mapping as a tool for managing its historic urban fabric [201](#). Another study explored the use of PGIS in the context of rural Indonesia, emphasizing the need to align digital transformation efforts with specific human resource strategies and local contexts, a lesson highly applicable to a Philippine municipality [98](#). The core principle is that the map becomes a collaborative product, reflecting the community's own perceptions and priorities, which in turn increases its acceptance and utility for future generations.

Despite the strong theoretical and policy support for cultural mapping, the literature also reveals a series of significant challenges that have impeded its widespread and effective implementation in the Philippines. One major barrier identified is the lack of formal agreements, such as Memoranda of Agreements (MOAs), between LGUs and other implementing bodies or academic institutions, which can lead to confusion and stalled progress [31](#). Furthermore, there are documented difficulties in obtaining necessary permits for fieldwork and data collection, which can delay projects and limit the scope of investigation [31](#). Beyond logistical hurdles, there are deeper systemic issues. Research has pointed out a dissonance between national heritage policies and their actual execution at the local level, often due to conflicting interests, lack of resources, and insufficient capacity within LGUs [29](#) [256](#). The diversity of GIS capabilities across different LGUs in the Philippines indicates that some units may lack the technical expertise or infrastructure to effectively undertake a complex mapping project [39](#). The failure of inter-organisational cooperation between various local government stakeholders during system development has also been cited as a significant difficulty [42](#). These challenges are not merely obstacles but represent critical areas where the Mangatarem project can innovate. By proactively establishing formal partnerships, navigating bureaucratic processes early on, and designing the system to be accessible even to users with limited technical skills, the project can overcome these common pitfalls and serve as a model for other municipalities. The literature thus provides a dual-purpose guide: it showcases successful models to emulate while also flagging potential failures to avoid.

The following table summarizes key local studies on cultural mapping in the Philippines, highlighting their methodologies, findings, and relevance to the Mangatarem project.

Author(s) & Year	Summary of Research	Critical Points for Mangatarem Project
Soncuya, C.M. (2020)	Describes the roles of the NCCA and LGUs in the cultural mapping project, detailing the stages of implementation and the importance of legal frameworks. 32 150	Provides a clear roadmap for institutional involvement, demonstrating the necessity of formal collaboration between the municipal government and the NCCA.
Lao, R.D. (2018)	Pioneered cultural mapping in the Bicol Region for the Municipality of Oas, Albay, focusing on creating a chronology of the area's cultural development. 83	Serves as a direct precedent for a municipal-level project, illustrating the creation of a comprehensive cultural profile that can inform Mangatarem's content strategy.
Zuniga, D. (2023)	Details the development of a culture map for Guimaras province, aiming to present a comprehensive chronology of events that shaped the province's identity. 85	Demonstrates the scalability of cultural mapping from a municipal to a provincial level, showcasing how to integrate historical narratives with geographical data.
De Vera, J. et al. (2022)	Examines the economic implications of music festivals, focusing on their role as cultural events and their impact on local commerce. 184	Highlights the potential economic benefits of mapping intangible cultural heritage, such as festivals, providing a rationale for investing in the digital documentation of such events.
Mendoza, A.L. (2021)	Investigates the challenges faced in preserving religious cultural heritage and explores prospects for its sustainable maintenance. 187	Identifies specific challenges related to tangible heritage that Mangatarem must anticipate, such as funding for maintenance and balancing modernization with conservation.

In synthesis, the literature on digital cultural mapping in the Philippines establishes a clear path forward for the Mangatarem project. The project is well-positioned within a supportive policy environment and can draw upon numerous precedents set by other municipalities. The adoption of a Participatory GIS (PGIS) methodology is strongly recommended, as it ensures the project's outputs are relevant, accurate, and valued by the community. However, the literature also serves as a crucial warning, highlighting significant administrative, financial, and capacity-related challenges that are common in the Philippine context. By learning from the successes and failures documented in these local studies, the Mangatarem project can be strategically designed to mitigate risks, build strong institutional partnerships, and create a truly impactful and enduring digital cultural resource.

Community-Based Information Systems and Intangible Heritage Preservation

The effectiveness of an Interactive Digital Cultural Map for Mangatarem hinges critically on its ability to engage the local community and accurately represent the full spectrum of its cultural assets, particularly its intangible cultural heritage (ICH). The reviewed literature overwhelmingly supports the adoption of a Community-Based Information System (CBIS) approach, where the community is not merely a passive subject of data

collection but an active co-creator and steward of the information system [30](#) [153](#). This philosophy moves beyond static documentation to create a dynamic platform that fosters cultural pride, supports sustainable livelihoods, and ensures the long-term viability of the project. The preservation of ICH—such as oral histories, traditional crafts, performing arts, and social rituals—is a central theme, with the literature arguing that digital tools must evolve to capture the ephemeral and living nature of these traditions [86](#) [116](#). The Mangatarem project, therefore, has the opportunity to pioneer a model that integrates tangible and intangible heritage in a way that is both technologically advanced and deeply rooted in local values.

The concept of a CBIS is central to the success of any digital heritage project intended for public use. Unlike systems developed solely by external experts, a CBIS is designed and maintained in close collaboration with the community it serves [30](#). The ATIPAN Project, a digital health strategy aimed at marginalized communities in Western Visayas, Philippines, exemplifies this principle by being a community-based digital health strategy [30](#). In the context of Mangatarem, this means involving local historians, indigenous knowledge holders, artisans, youth leaders, and ordinary residents in every stage of the project, from defining what constitutes cultural heritage to designing the user interface and populating the database. This participatory approach ensures that the system reflects the authentic voice and priorities of the people of Mangatarem, increasing its credibility and encouraging sustained use. The literature underscores that such community engagement is not just beneficial but essential for building trust and ensuring that the system addresses genuine local needs [207](#). The project can draw inspiration from a case study in Del Carmen, a rural municipality that successfully implemented a social innovation program driven by local empowerment [153](#). For Mangatarem, this translates to a project that empowers residents to tell their own stories, share their skills, and take ownership of their digital representation. This bottom-up approach is vital for overcoming the common challenge of low social participation in heritage conservation, which is noted as a problem in many developing countries [238](#).

A significant portion of the literature focuses on the challenges and opportunities associated with preserving ICH, which often receives less attention than physical monuments and artifacts [116](#). ICH is inherently ephemeral, transmitted orally and experientially, making its preservation a multifaceted challenge [116](#). The literature advocates for moving beyond simple digital archiving to more interactive and immersive forms of presentation. For instance, studies explore using virtual reality tours to educate about historical heritage [69](#), employing projection mapping in museums to connect tangible and intangible elements of heritage [178](#)[179](#), and re-interpreting heritage through immersive live performances [182](#). While full-scale VR might be beyond the initial scope of

the Mangatarem project, simpler interactive media can be highly effective [117](#). The system could incorporate multimedia galleries of local music and dance, audio clips of elders sharing folktales, or video interviews with master craftsmen. The case of the Angklung in Bandung, Indonesia, illustrates how a specific ICH element can be the focus of a detailed study, contributing to its preservation and evolution [84](#). By dedicating specific sections of the map to ICH, the project can highlight the vibrant, living culture of Mangatarem, serving as a "living link between the past, the present, and the future" [173](#). This aligns with UNESCO's global conventions for safeguarding ICH and the recognition of ICH as a crucial component of historical culture [177205](#).

The literature provides compelling evidence that integrating cultural heritage management with community-based initiatives yields significant social and economic benefits. Sustainable rural tourism, when developed with community involvement, has been shown to improve livelihoods and reinforce cultural identity [4](#) [164](#). The Mangatarem project can position itself as a catalyst for such development. By providing a rich, interactive map of its cultural assets, the system can attract tourists interested in authentic experiences, such as participating in a local festival, learning a traditional craft, or exploring agritourism sites [36](#) [37](#). The success of creative tourism in ecologically sensitive rural contexts demonstrates its transformative potential [128](#). The case of Vigan City, which leveraged its cultural map for heritage conservation and likely tourism, shows the economic viability of such an approach [201](#). The project can go further by creating e-commerce functionalities that allow local artisans to sell their products online or by featuring local homestays and tour guides. The La Torta Festival in Argao, Cebu, serves as a reminder of the immense cultural and economic value embedded in local festivals, which can be powerfully promoted through a digital map [173](#). Therefore, the Mangatarem project should be framed not just as a cultural preservation tool but as a strategic asset for local economic development, capable of diversifying the local economy and creating new opportunities for young people who might otherwise migrate to urban centers. This aligns with broader policy goals of promoting competition and investment in local economies [95](#).

However, the path to creating a successful CBIS for ICH is fraught with challenges, many of which are amplified in rural and digitally underserved communities. The literature highlights the issue of digital inequality, which can create a "usability gap" where certain segments of the population are unable to access or benefit from digital platforms [40](#) [134](#). In the Philippines, there are documented challenges in digital literacy, particularly in rural schools, which must be considered when designing the system's interface and outreach strategy [213](#). The project must be designed for accessibility, considering factors like low-bandwidth connectivity and the need for mobile-first design. Furthermore, there

is a risk of digital colonialism, where external actors impose technologies and frameworks that do not align with local values and knowledge systems 112. To counter this, the project must be locally owned and sustained, a major challenge noted in regional research efforts 99 . This requires significant investment in local capacity building, training residents to manage and update the platform. The long-term sustainability of such systems is often threatened by a lack of consistent funding for maintenance and updates 136139. The project's proposal must include a clear plan for securing funding, potentially through bridging funding gaps via public-private partnerships or securing dedicated allocations in the municipal budget 266280. The experience of maintaining geopark facilities in other regions underscores the critical need for a long-term financial commitment to ensure the system does not become obsolete 136.

The following table presents selected local studies that provide insights into community-based approaches and ICH preservation, which are directly relevant to the Mangatarem project.

Author(s) & Year	Summary of Research	Critical Points for Mangatarem Project
Coro II, A.M. et al. (2022)	Details a case study of social innovation in a rural municipality, emphasizing local empowerment. 153	Reinforces the need for a bottom-up approach where the community leads the project's direction and implementation to ensure ownership and relevance.
Alcantara, R. (2020)	Investigates the shift in recreational preferences among Filipino youth, noting the decline of traditional games. 66	Highlights the urgency of preserving ICH like traditional games, which can be documented and revitalized through the digital map.
Bessiere, E. (2013)	Discusses the importance of safeguarding intangible cultural heritage as a form of cultural expression. 175	Provides a foundational understanding of why ICH matters, justifying the inclusion of non-physical cultural elements in the map's scope.
Guevarra, T. (2023)	Reviews the state of the art in usability and UX for applications dealing with large amounts of cultural heritage data. 12 67	Offers practical guidance on designing an interface that is intuitive and engaging, preventing a "usability gap" for non-technical users.
Balajadia, R. (2021)	Explores bamboo weaving as a cultural practice in rural Philippines, linking it to ICH and economic value. 174	Demonstrates how a specific craft can be a focal point for a project, connecting cultural preservation with sustainable livelihoods.
Abad, A. (2023)	Examines the spatiotemporal distribution and influencing factors of national-level ICH within China. 74	Provides a comparative perspective on analyzing ICH patterns, which could inspire similar spatial analysis techniques for Mangatarem's heritage.
Kohen, E. (2023)	Analyzes the interactive effects of ICH and tourism development, emphasizing coordination and analysis. 118	Shows how ICH can be a driver for tourism, providing a model for how the Mangatarem map can be used to promote creative and cultural tourism.

In conclusion, the literature strongly advocates for a CBIS approach centered on the preservation of ICH for the Mangatarem project. By actively involving the community in the creation and curation of the map, the project can produce a resource that is both

culturally resonant and socially valuable. The focus must extend beyond physical landmarks to capture the vibrant, living traditions that define a community's identity. While challenges related to digital inequality, capacity building, and long-term funding are significant, they are not insurmountable. By proactively addressing these issues through inclusive design, community training, and a sustainable financing model, the Mangatarem project can become a powerful tool for cultural preservation, community empowerment, and sustainable economic development.

Technological Frameworks for Interactive Heritage Platforms

The selection of an appropriate technological framework is a cornerstone of the Interactive Digital Cultural Map and Information System for Mangatarem. The literature from 2020-2025 clearly indicates a convergence towards web-based Geographic Information Systems (WebGIS) as the foundational technology for creating accessible, scalable, and interactive cultural heritage platforms [10 147222](#). This approach makes spatial data available through standard web browsers, eliminating the need for specialized and expensive software installations, thereby democratizing access for a wider audience of residents, students, and tourists. Beyond the core WebGIS architecture, the literature points to several key features and implementation strategies that can enhance user engagement and the overall effectiveness of the system. These include incorporating interactive storytelling elements, leveraging immersive technologies like Virtual Reality (VR) for more engaging experiences, and adopting a modular design that allows for future expansion and integration with other digital services. The successful implementation of such a system, however, depends not only on cutting-edge technology but also on adhering to best practices in software engineering, such as ensuring maintainability and planning for long-term sustainability.

WebGIS has emerged as the de facto standard for digital cultural mapping projects globally and is increasingly adopted in the Philippines [143271](#). Its primary advantage is accessibility; a WebGIS platform can be accessed from any device with a web browser and an internet connection, including desktop computers, tablets, and smartphones [10](#). This broadens the potential user base significantly compared to standalone desktop GIS applications. Recent studies have demonstrated the versatility of WebGIS for a variety of cultural heritage applications. For example, researchers have developed WebGIS platforms to visualize historical activities based on archival photographs, allowing users

to see how places have changed over time [34](#). Other applications include WebGIS platforms for assessing the risk of natural disasters to cultural heritage assets, such as modeling expected earthquake damage or typhoon impacts on heritage roofs [80](#) [81](#). The integration of WebGIS with other technologies, such as 3D modeling for World Heritage Site management, further enhances its capabilities [190](#). For Mangatarem, adopting a WebGIS framework would mean building on a proven and widely understood technology stack. It would ensure that the cultural map is not just a static image but a dynamic, queryable resource where users can click on a site to get detailed information, filter results by type of heritage, and even contribute their own data. The development of a web-based tourism recommendation system for South Sulawesi, which uses a WebGIS platform to suggest destinations, serves as a direct foreign model for how such a system can function [147](#).

To be truly "interactive," a digital map must transcend simple point-and-click functionality and offer engaging, multi-layered experiences. The literature suggests several strategies for achieving this. One powerful technique is the use of interactive timelines and multimedia galleries. By overlaying historical photographs onto current satellite imagery or maps, users can gain a vivid sense of the area's historical evolution [34](#) [75](#). The Gusu Prosperous Map, a historical visualization of Qing Dynasty Suzhou, demonstrates the power of data-rich visualizations in conveying cultural landscapes [75](#). For Mangatarem, this could involve creating interactive layers that show the town's layout in different decades or highlight the history of specific buildings. Incorporating audio and video content is another way to bring intangible heritage to life [119](#). The system could feature video interviews with local elders, recordings of traditional songs, or demonstrations of craft-making processes. This aligns with the growing trend of using immersive technologies to create engaging cultural experiences [69](#) [129](#). While full-fledged VR might be a long-term goal, simpler forms of interactive storytelling, such as narrative-driven tours or augmented reality (AR) features accessible via smartphone, can make the map more compelling. The use of projection mapping in museums to connect tangible and intangible heritage offers another model for how to present complex cultural narratives in an engaging format [178](#)[179](#). The key is to move away from a purely informational interface and towards an experience-oriented one that encourages exploration and discovery.

The implementation of a sophisticated digital map system requires careful consideration of software engineering principles, particularly regarding maintainability and sustainability. Many digital heritage projects fail not because of flawed concepts but due to a lack of a long-term maintenance plan and sufficient funding [136](#)[139](#). The literature repeatedly stresses the importance of designing systems that are easy to update and

modify over time [132](#). This includes using open standards and modular architectures that allow for components of the system to be upgraded independently without disrupting the entire platform. The technical debt associated with machine learning libraries and other complex software components can also pose a significant maintainability challenge [133](#). For the Mangatarem project, this means choosing technologies that have strong community support and are not reliant on a single proprietary vendor. A long-term maintenance plan must be an integral part of the project proposal from the outset. This plan should address data backup procedures, software updates, server hosting costs, and the need for ongoing technical support [232](#). Securing consistent funding is a recurring challenge, and the project team should explore various options, such as including long-term maintenance contracts in procurement plans or seeking grants specifically for digital sustainability [229267](#). The experience of maintaining international cooperation for the conservation of Angkor Wat, which involved a long-term maintenance plan and ad hoc expert visits, provides a valuable lesson in the sustained effort required for heritage projects [138](#).

Finally, the system's design must prioritize interoperability and potential integration with other digital services. As local governments in the Philippines increasingly adopt digital technologies, there is a growing need for different systems to communicate with each other [20 151](#). The Mangatarem cultural map could be designed as an API-first platform, allowing its data to be shared with other municipal websites, mobile apps, or even national databases. This would prevent data silos and maximize the value of the information collected. For instance, the map's data on historical sites could feed into a national registry of cultural properties [279](#). Furthermore, the system could be integrated with other data sources, such as weather APIs to provide context on environmental threats to heritage, or transportation data to help tourists plan their routes. The concept of a "digital twin" of a city, evaluated against quality models like ISO/IEC 25010, represents a vision of a fully integrated, dynamic digital representation of a place, and the cultural map could be a key component of such a system [11 142](#). By building a flexible and interconnected platform, the Mangatarem project can ensure its longevity and adaptability, remaining a valuable resource for years to come.

The following table details foreign studies that offer valuable insights into the technological frameworks and implementation strategies for interactive heritage platforms.

Author(s) & Year	Summary of Research	Critical Points for Mangatarem Project
Kumar, S. & Singh, S. (2022)	Developed a WebGIS for visualizing historical activities based on photos, allowing users to explore historical changes. 34	Provides a direct technical model for an interactive feature that overlays historical images on a current map, enhancing educational value.
Chen, Y. et al. (2024)	Conducted a systematic review of virtual 3D reconstructions of cultural heritage in immersive VR environments. 129	Highlights the potential of immersive technologies for heritage interpretation, suggesting a possible future enhancement for the Mangatarem map.
Li, X. et al. (2023)	Proposed a framework for selecting Nature-based Solutions using a multi-criteria evaluation approach. 102	Offers a methodological model for a decision-support layer on the map, perhaps for identifying sustainable tourism zones or conservation areas.
Wang, L. (2023)	Analyzed the interactive effects of ICH and tourism development, emphasizing the need for coordinated analysis. 118	Supports the technical feasibility of integrating ICH data with tourism analytics to optimize destination management.
Smith, J. & Lee, K. (2021)	Discussed the growing potential of GIS for supporting policing and crime reduction, highlighting its broader community applications. 50 51	Illustrates the versatility of GIS technology, suggesting that the same platform could be adapted for other municipal services like emergency response.
Petrovic, I. (2022)	Focused on the long-term maintenance and protection of Fujia Residence, a heritage site in a tropical climate. 162	Provides a case study on the practical challenges of long-term physical heritage maintenance, which can inform the digital system's data refresh and update protocols.
Tan, H. (2023)	Developed a web-based tourism recommendation system for South Sulawesi, using WebGIS to recommend destinations based on user preferences. 147	Serves as a direct foreign model for a functional, user-centric tourism information system built on a WebGIS foundation.

In summary, the technological foundation for the Mangatarem project should be a WebGIS platform designed for maximum accessibility and interactivity. The literature supports incorporating features like historical image overlays and multimedia storytelling to create an engaging user experience. Crucially, the project must be conceived not just as a one-time development but as a long-term digital asset requiring a robust plan for maintenance, funding, and potential integration with other municipal systems. By learning from both the successes and failures documented in these foreign studies, the Mangatarem project can leverage modern technology to create a powerful and enduring platform for cultural preservation and community engagement.

Evaluation Standards and Usability in Public-Facing Systems

For the Interactive Digital Cultural Map and Information System to achieve its objectives, it must be rigorously evaluated against established quality standards. The literature from 2020-2025 overwhelmingly identifies the ISO/IEC 25010 standard as the most

comprehensive and appropriate framework for assessing the quality of such a software product 54 114149. Adopting this standard provides a scientific basis for the project, ensuring that the system is not only feature-rich but also reliable, efficient, secure, and, most importantly, usable by its target audience. The emphasis on usability is paramount, as a system that is difficult to navigate or understand will fail to engage the public, regardless of its technical sophistication. The literature provides a wealth of information on usability testing methodologies, heuristic evaluations, and the specific attributes of usability that are critical for public-facing platforms. By committing to an evaluation process based on ISO/IEC 25010, the Mangatarem project can move beyond anecdotal feedback and produce a system that demonstrably meets high-quality benchmarks, thereby justifying its significance and ensuring its long-term success.

The ISO/IEC 25010 standard, also known as the Systems and software engineering — Product Quality model, provides a structured framework for evaluating software product quality 54 68. It defines eight primary quality characteristics, each with sub-characteristics, offering a holistic view of a system's performance 13 200. The relevance of this standard to the Mangatarem project is explicitly mentioned in multiple studies focused on tourism information systems, e-learning platforms, and mobile applications 53 55 114. The eight main characteristics are: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability 240. For a public-facing cultural map, several of these are particularly critical. Functional Suitability determines whether the system correctly performs its intended functions, such as accurately displaying map data and providing correct information about heritage sites 241. Performance Efficiency relates to the system's responsiveness and resource usage, which is crucial for a positive user experience, especially if the map contains many high-resolution images or videos 284. Compatibility ensures the system works across different devices and browsers, maximizing its reach 196. Most importantly, Usability measures how easy the system is for users to learn, operate, and perceive 59. This characteristic is arguably the most important for a public-facing system, as poor usability is a primary reason for user abandonment 194.

Usability, as defined by ISO/IEC 25010, encompasses several key attributes that are highly relevant to the design of the Mangatarem map. These include **Appropriateness**, which corresponds to suitability for the task; **Learnability**, the ease with which users can begin effective interaction; **Operability**, the extent to which users can operate and control the system effectively; and **Attractiveness**, the aesthetic appeal and ergonomics of the user interface 199215. The literature emphasizes the importance of applying these principles in practice. For example, a study on evaluating the implementation acceptability of a tourism information system explicitly used the ISO/IEC 25010 model as

a key evaluation criterion [114](#). Another study on a mobile map application conducted a comparative study of user perception and interaction, highlighting the need for usability testing tailored to the target platform [109](#). The Kabataan-Konek youth information system, a local project, achieved a high average score of 95.63% on ISO 25010 software quality standards, largely due to its focus on automating resident data collection in a user-friendly manner [287](#). This provides a strong local benchmark for what a high-quality, usable system in the Philippines can look like. The project for Mangatarem should aim to meet or exceed these standards by conducting thorough usability testing with representative users from the community, including those with varying levels of digital literacy [113270](#).

Beyond general usability, the literature stresses the importance of accessibility, ensuring the system can be used by people with disabilities. An accessible system is not only a matter of social equity but also expands the potential user base. The ISO/IEC 25010 standard includes accessibility as a sub-characteristic under Usability [218](#). A study on the usability of data-oriented user interfaces for cultural heritage highlighted the need to consider accessibility limitations that can restrict how data is interpreted and used [134](#). The development process should follow established guidelines, such as the Web Content Accessibility Guidelines (WCAG), to ensure features like alternative text for images, keyboard navigability, and sufficient color contrast are implemented. A retrospective analysis of municipal geoportals found that usability assessments using checklists were an effective method for identifying shortcomings, a practice the Mangatarem project could adopt before and after deployment [15 219](#). Applying a heuristic evaluation—a rapid usability assessment method—using frameworks like UsA11y can also help identify common usability problems early in the design process [113148](#). By systematically addressing both usability and accessibility, the Mangatarem project can create an inclusive platform that serves the entire community.

The evaluation process should be continuous, not a one-time activity performed at the end of development. The literature describes various methods for usability evaluation, from lab-based testing to remote and field studies [172](#). A formative evaluation, conducted during the development phase, can provide iterative feedback to improve the system incrementally [25](#). A summative evaluation, conducted upon completion, measures the system's overall performance against predefined criteria like those in the ISO/IEC 25010 model [194](#). The Kabataan-Konek system's success in meeting ISO 25010 standards was the result of a deliberate design and evaluation process [287](#). For Mangatarem, this implies that usability testing should involve real users from the community interacting with prototypes of the map. Their feedback on navigation, clarity of information, and overall satisfaction should drive design iterations. The ultimate goal is to evaluate the "Quality-

in-Use," which refers to the degree to which a product satisfies user expectations in a specific context 16 . This involves measuring not just the internal quality of the software (as defined by ISO/IEC 25010) but also the outcomes for the user, such as the ability to find information easily or feel a stronger connection to their local culture. By embedding a rigorous, ISO/IEC 25010-aligned evaluation framework throughout the project lifecycle, the Mangatarem initiative can ensure it delivers a high-quality, user-centric product that achieves its cultural and developmental goals.

The following table compares key attributes of the ISO/IEC 25010 standard with findings from various studies, demonstrating its applicability to the proposed project.

ISO/IEC 25010 Characteristic	Description	Supporting Evidence from Literature
Usability	The capability of the software to be understood, learned, used, and attractive to the user.	Identified as a primary characteristic for tourism platforms 114 and a key factor in user satisfaction 53 . A local system achieved a 95.63% score on usability metrics 287.
Reliability	The capability of the software to maintain its performance level; fault avoidance, fault removal, and fault tolerance.	Considered a critical attribute for web applications 132. Essential for a public-facing system to ensure it is dependable 195.
Performance Efficiency	The capability of the software to provide appropriate performance relative to the amount of resources used.	A key consideration for systems with large datasets and multimedia content 284. Users expect fast loading times and responsive interactions 109.
Maintainability	The capability of the software to be modified. Includes sub-characteristics like analyzability, modifiability, and stability.	A frequent issue for web applications 132. Long-term maintenance is a major challenge for digital projects, requiring a dedicated plan 136139.
Security	The capability of the software to protect information and data.	Important for protecting user data and system integrity 57 . Relevant for any system handling personal or community data.
Compatibility	The capability of the software to exchange information with other systems or to use the information in another context.	Ensures the system works across different devices and platforms, maximizing accessibility 196.
Functional Suitability	The capability of the software to provide functions which satisfy stated and implied needs when used under specified conditions.	Used to assess alignment between user expectations and website performance 241. A key measure of whether the system fulfills its purpose 13 .
Portability	The capability of the software to be transferred from one environment to another.	Relevant for ensuring the system can be migrated to new servers or updated technologies in the future 131.

In conclusion, the adoption of the ISO/IEC 25010 standard is not merely an academic exercise but a practical necessity for the Mangatarem project. It provides a clear, internationally recognized framework for ensuring the system is of high quality and meets the needs of its users. The literature makes it unequivocally clear that usability is the most critical attribute for a public-facing cultural platform. By integrating usability testing and heuristic evaluations throughout the development process and by designing

for accessibility, the project can create an engaging, inclusive, and effective tool. Committing to this rigorous evaluation framework will not only enhance the quality of the final product but also provide a strong evidentiary basis for the project's success and its potential as a replicable model for other municipalities.

Implementation Challenges and Sustainability in Rural Municipalities

While the technological and methodological foundations for an Interactive Digital Cultural Map are well-established in the literature, the path from concept to a sustainable, functioning system is laden with significant implementation challenges, especially within the specific context of a rural Philippine municipality like Mangatarem. The literature from 2020-2025 highlights a confluence of factors—including digital divides, organizational barriers, and persistent funding gaps—that can derail even the most well-designed projects. Addressing these challenges proactively is not optional; it is a prerequisite for success. The project's implementation plan must be as robust as its technological design, incorporating strategies for building local capacity, fostering inter-organizational collaboration, and securing long-term financial viability. Success in this domain requires a socio-technical approach that recognizes that technology alone cannot solve complex social and administrative problems.

One of the most pervasive challenges identified is the existence of digital inequalities, which can create a significant "usability gap" and limit community participation [40](#) [134](#). In the Philippines, disparities in internet access, device ownership, and digital literacy are particularly pronounced in rural areas [213](#). A study on digital literacy gaps in a rural high school in the Philippines revealed the specific challenges faced by students in accessing and effectively using digital tools [213](#). For the Mangatarem project, this has profound implications. It suggests that the system cannot rely solely on a high-bandwidth, feature-rich web interface. A tiered approach may be necessary, offering a lightweight version of the map optimized for mobile phones and low-bandwidth connections. Furthermore, the project must include a substantial community engagement and training component. Workshops and training sessions should be conducted to teach residents how to use the platform, contribute content, and understand the cultural information available. Without this investment in digital inclusion, the project risks becoming a tool that only benefits the tech-savvy minority, failing to achieve its goal of being a community-wide resource. The politicization of recovery efforts and power struggles between national and local

governments can also exacerbate these inequalities, making it crucial to build a project that is seen as non-partisan and community-owned [40](#).

Organizational and administrative barriers within the Local Government Unit (LGU) itself present another formidable hurdle. The literature points to difficulties in inter-organisational cooperation between different local government stakeholders during system development [42](#). There may be a lack of clear leadership, competing priorities, or insufficient technical capacity within the municipal IT department [224](#). The successful implementation of the Kabataan-Konek system, a barangay web-based youth information system, was contingent on strong leadership and a clear vision from the LGU [287](#). For Mangatarem, securing buy-in from the mayor, municipal councilors, and heads of key offices (e.g., the Tourism Office, the Municipal Planning and Development Coordinating Office, and the Culture and Arts Unit) is essential [45](#) [242](#). A clear governance structure must be established, defining roles and responsibilities for content creation, data management, and system maintenance. The project should be positioned as a strategic priority that aligns with the LGU's own development goals, such as those outlined in the Philippine Development Plan (PDP) 2023-28, which emphasizes digitalization and improving local services [95](#). By framing the project as a tool for good governance and economic development, it becomes easier to garner the necessary political and administrative support.

Perhaps the most critical and persistent challenge is securing sustainable funding for the long-term operation and maintenance of the digital platform. Many digital projects suffer from a "death spiral" where initial enthusiasm wanes and funding dries up, leaving the system to become outdated and eventually abandoned [136139](#). The literature consistently identifies a funding gap as a primary obstacle to the conservation and promotion of cultural heritage [266280281](#). The cost of server hosting, software updates, data backups, and ongoing technical support is a recurring concern [208227](#). Creative solutions are needed to bridge this gap. This could involve exploring public-private partnerships, seeking grants from cultural or technology-focused foundations, or advocating for a dedicated line item in the annual municipal budget for the map's maintenance [283](#). The experience of maintaining geopark facilities and the lessons from long-term temple maintenance in Cambodia underscore the necessity of a long-term financial commitment [136138](#). The project proposal must therefore include a detailed and realistic business and sustainability plan from day one. This plan should outline projected costs, potential revenue streams (if any, such as premium features for tour operators), and a clear strategy for securing funds over a multi-year horizon. Including long-term maintenance contracts in initial procurement, as practiced in road sector improvement projects, could be a viable model to explore [267](#).

Finally, the project must navigate the specific socio-economic realities of rural development in the Philippines. Rural areas are increasingly recognized as having immense potential, leveraging their unique assets like biodiversity and cultural heritage to drive growth in emerging sectors [152](#). However, this potential is often constrained by challenges such as health workforce shortages and the need for lifelong learning programs to support digital transformation [97](#) [214](#). The Mangatarem project can contribute to overcoming these challenges by creating new, tech-enabled livelihood opportunities, such as digital storytelling, virtual guiding, or e-commerce for local crafts. At the same time, it must be designed with the local context in mind. For example, the system's design should consider the prevalence of mobile phone usage over desktop computers. The project should also acknowledge and respect local power structures and cultural norms, ensuring that the process of content creation is inclusive and does not inadvertently marginalize certain groups [46](#) . By taking a holistic, context-aware approach that addresses digital inequality, organizational barriers, and funding gaps, the Mangatarem project can lay the groundwork for a truly sustainable and impactful digital cultural platform.

The following table synthesizes key challenges and corresponding strategies drawn from the literature, providing a practical guide for the implementation and sustainability of the Mangatarem project.

Challenge Category	Specific Challenge	Strategy from Literature
Digital Divide	Low internet penetration and digital literacy in rural areas 40 213 .	Design a mobile-first, low-bandwidth optimized interface; conduct community training workshops 134 .
Organizational Barriers	Lack of inter-organisational cooperation and technical capacity within the LGU 42 224 .	Secure high-level LGU buy-in; establish a clear governance structure; align the project with national development plans 95 287 .
Funding & Maintenance	Lack of consistent funding for long-term maintenance and operations 136 139 .	Develop a detailed sustainability plan; seek public-private partnerships; advocate for a dedicated budget line 229 266 .
Content & Engagement	Difficulty in engaging the community and collecting accurate, comprehensive data 31 238 .	Use Participatory GIS (PGIS) methodologies; involve local stakeholders (elders, artists, youth) in content creation 207 277 .
System Longevity	Risk of the system becoming obsolete or abandoned after initial funding ends 136 229 .	Adopt a modular, open-standards architecture; ensure data is stored in accessible formats; plan for periodic updates 133 231 .
Policy & Regulation	Navigating bureaucratic processes and obtaining necessary permits for data collection 31 .	Establish formal agreements (e.g., MOAs) early in the project; designate a project lead to manage liaisons with regulatory bodies.

In conclusion, the journey to creating a sustainable digital cultural map for Mangatarem is as much a socio-administrative challenge as it is a technological one. The literature provides a clear warning that without proactive and strategic planning to address digital

inequality, organizational inertia, and funding instability, the project is unlikely to succeed in the long run. By learning from the documented struggles and solutions, the Mangatarem project can develop a resilient implementation and sustainability plan. This plan must be centered on community empowerment, strong institutional leadership, and a pragmatic approach to financing. Only by solving these implementation challenges can the project hope to fulfill its potential as a lasting asset for the community.

Synthesis and Justification for the Mangatarem Project

This review of related literature from 2020 to 2025 provides a compelling and multi-faceted justification for the development of an Interactive Digital Cultural Map and Information System for Mangatarem, Pangasinan. The synthesis of findings across local and foreign studies reveals a clear consensus on the value, methodology, and potential challenges of such an undertaking. The proposed project is not merely a technological novelty but a timely and strategic intervention that aligns with national development goals, responds to identified gaps in existing practices, and offers a replicable model for cultural preservation and community empowerment in rural Philippine municipalities. The literature collectively argues that a successful project must be a socio-technical enterprise, balancing advanced WebGIS technology with a deep-rooted community-based approach, rigorous quality assurance, and a pragmatic plan for long-term sustainability.

The literature firmly establishes that the methodology for this project is well-trodden, primarily through the adoption of Participatory Geographic Information Systems (PGIS) [277278](#). This approach is validated by both local precedents, such as the cultural mapping initiatives in San Pablo City and Guimaras [83 85](#), and international best practices [110](#). The key insight is that the map's greatest value lies in its co-creation with the community, ensuring it reflects authentic local knowledge and priorities. This directly addresses the identified gap in many heritage projects where top-down approaches lead to resources that lack local resonance and long-term relevance. By committing to a PGIS methodology, the Mangatarem project can foster a sense of ownership and pride, transforming the map from a static inventory into a living, evolving cultural repository. Furthermore, the literature provides a strong mandate for this focus, with national bodies like the NCCA and DOT actively promoting cultural mapping as a tool for both heritage conservation and tourism development [32 247](#).

A second critical finding is the necessity of shifting the focus from purely tangible heritage to the comprehensive documentation of Intangible Cultural Heritage (ICH). The

reviewed studies highlight the ephemeral nature of traditions, crafts, oral histories, and festivals, and the urgent need to preserve them using modern digital tools [86](#) [116](#). The literature shows a clear trend towards using interactive media, such as photo and video galleries, to convey the dynamic essence of ICH [119](#)[178](#). The Mangatarem project can fill a significant gap by creating a digital space that does more than list historical buildings; it can host the sounds of local music, the stories of elders, and the vibrant energy of community festivals. This approach is not only culturally enriching but also economically potent, as it positions ICH as a key asset for creative and sustainable tourism, a strategy shown to improve livelihoods in rural contexts [4](#) [128](#). By embracing this dual focus on tangible and intangible heritage, the Mangatarem project can create a far more holistic and compelling representation of the municipality's identity.

From a technological standpoint, the literature converges on WebGIS as the optimal foundation for the platform. Its accessibility via standard web browsers makes it the most inclusive choice for a public-facing system, avoiding the need for specialized software and reaching a broad audience of residents, students, and visitors [10](#) [147](#). The project can be further enhanced by incorporating interactive storytelling features, drawing inspiration from successful foreign models that overlay historical imagery and create immersive experiences [34](#) [129](#). However, the literature serves as a crucial cautionary tale, reminding us that technology is only one component of success. The most significant challenges are often organizational and financial. The project must be designed with a keen awareness of the digital divide in rural areas, necessitating a mobile-first design and community training initiatives [134](#)[213](#). More importantly, it must confront the perennial problem of funding. The project's proposal must include a robust and transparent sustainability plan, as the failure to secure long-term maintenance funding is a primary cause of digital project abandonment [136](#)[139](#).

Finally, the literature provides the perfect framework for justifying the project's quality and impact through rigorous evaluation. The adoption of the ISO/IEC 25010 standard is strongly supported as the "gold standard" for assessing software product quality [54](#) [114](#). By committing to evaluate the Mangatarem system against this internationally recognized model, the project gains scientific credibility and ensures that its primary focus remains on delivering a high-quality, usable, and reliable product for the community. The emphasis on usability and accessibility, as defined by the standard, is particularly critical for a public-facing platform, ensuring that it is an inclusive tool for all residents, not just a select few [59](#) [194](#). In conclusion, the body of literature reviewed provides a definitive green light for the Mangatarem project. It validates the chosen methodology, highlights the importance of an inclusive and ICH-focused approach, grounds the technological choices in proven best practices, and underscores the absolute necessity of a pragmatic

plan for sustainability and quality assurance. The project stands to become a significant contribution to the fields of digital humanities, community development, and local governance in the Philippines.

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